

A RESEARCH PROJECT ON FINANCIAL INFORMATION RETRIEVAL

FinAlign.

*Bridging the financial
vocabulary gap
between people and
documents.*

UNIVERSITÀ DI PAVIA
DI PILATO · CINQUEMANI · MARCHIONI
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THE PROBLEM

People and documents *don't speak the same language.*

Traditional search relies on literal keyword matching — and fails the non-experts who need financial information most.

USER ASKS	"Should I put money in property funds?"
DOC SAYS	"REIT allocation under current yield curve assumptions..."
USER ASKS	"Is this a good time to buy a house?"
DOC SAYS	"30-year fixed mortgage spreads vs. treasuries..."

“Does query rewriting, prefix-based encoding, and document chunking improve the effectiveness of a hybrid BM25 + dense retrieval system for financial questions?”

THE DATASET

FiQA, from the *BEIR benchmark*.

57,600 DOCUMENTS News headlines, microblog posts, and report snippets.	648 QUERIES Natural-language financial questions from non-experts.	1,705 QRELS Graded relevance annotations for evaluation.
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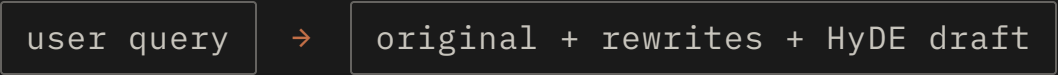
“What is the difference between a REIT and a real estate stock?”

THE PIPELINE

From a question to *a ranked list*.

01 Query Rewriting & *HyDE*

An LLM expands the user query with technical financial vocabulary and drafts a hypothetical answer.



02 Hybrid *BM25 + BGE*

BM25 retrieves on the lexical index. BGE retrieves on a chunked dense index, with chunk scores max-pooled to the document level. Both runs are fused (RRF or linear).



03 Document *Chunking*

Documents are split into overlapping passages before dense indexing — exposing localized relevance signals to the bi-encoder.

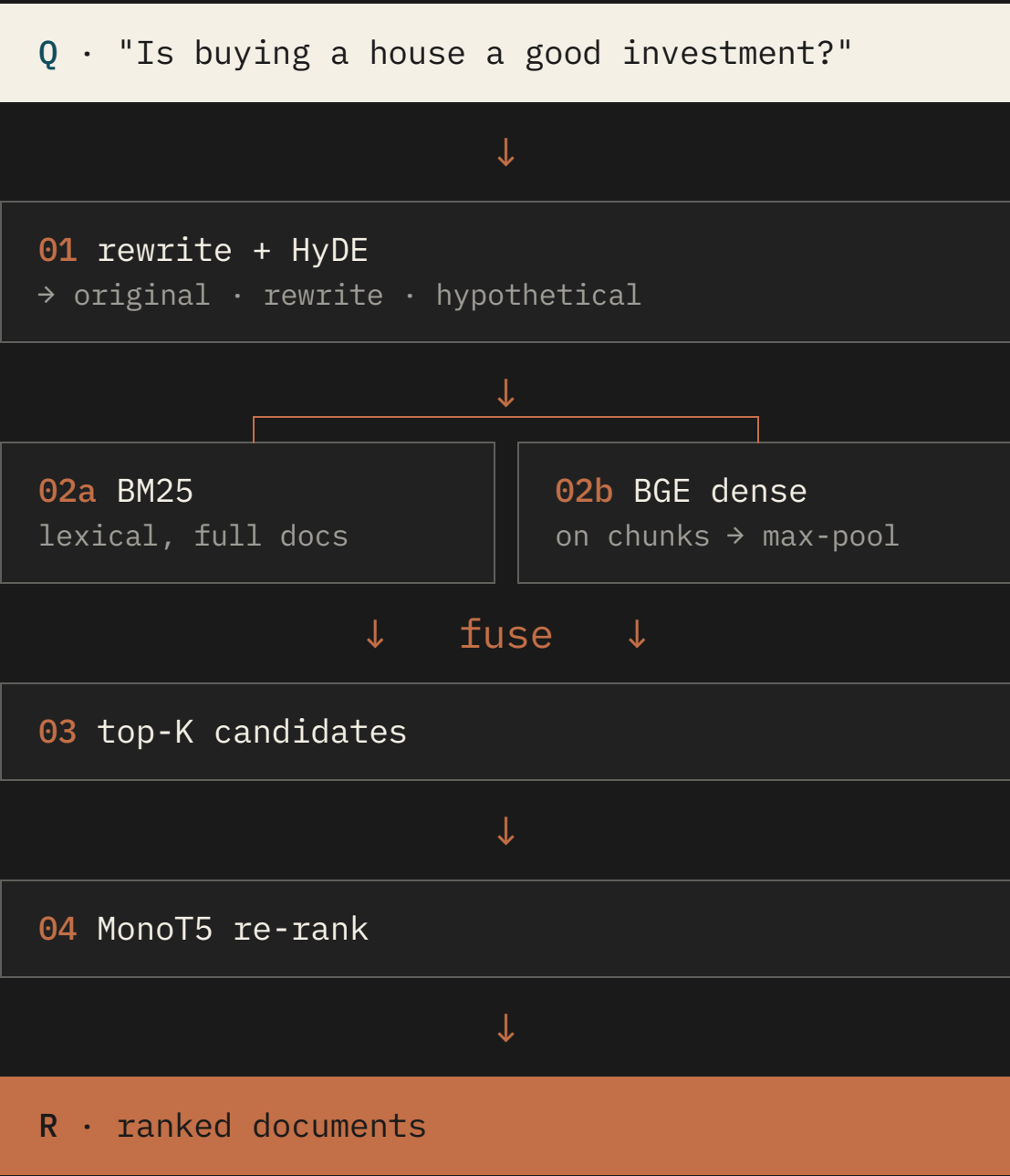


04 MonoT5 *Re-ranking*

A cross-encoder rescores the top candidates by jointly encoding query–document pairs.



DATA FLOW



One question, *many semantic views*.

An LLM rewrites the query with technical financial vocabulary, then drafts a *hypothetical answer* (HyDE) used as a semantic proxy.

Original, rewrites, and HyDE are fused via score combination — diversification, not substitution.

ORIGINAL

"Is buying a house a good investment right now?"

REWRITE

"Residential real estate purchase ROI under current mortgage rates and housing market conditions."

HYDE

"Buying a house can be a good investment depending on local price-to-rent ratios, mortgage spreads, and expected holding period..."

Two retrievers, *one ranking*.

SPARSE

BM25

First-stage lexical retrieval. Term saturation, document-length normalization, exact-match recall.

Index · Terrier inverted

Strength · Rare terms, exact matches

DENSE

BGE bi-encoder

Semantic retrieval over BGE embeddings. Captures paraphrases and implicit financial concepts beyond lexical overlap.

Index · FAISS vector store

Strength · Synonyms, intent, paraphrase

QUERY PREFIX

"Represent this sentence for searching relevant financial passages:"

↓ WHAT WE FOUND

FINALIGN

In our experiments, fusing BM25 with instruction-tuned BGE **did not improve over BGE alone**. The dense retriever already captured most of the signal — see slide 10.

STAGE 02 · SCORE FUSION (RRF · LINEAR)

DOCUMENT CHUNKING

Retrieve passages, *rank documents*.

Long documents dilute embeddings.
Splitting into overlapping fixed-length
passages exposes localized relevance
signals to the dense retriever.

Chunk scores are aggregated to the
document level using `max pooling`
— one strong passage promotes its
parent document.

DOCUMENT · SPLIT INTO CHUNKS

Introductory paragraph on portfolio diversification...

CHUNK 01

Direct discussion of REIT vs. real-estate stock structure, dividend treatment, and liquidity profile...

CHUNK 02

Tangential commentary on tax filing dates...

CHUNK 03

Closing remarks and disclaimer...

CHUNK 04

↓ max-pool over chunks ↓

Document score

= $\max(s_1, s_2, s_3, s_4)$

A cross-encoder for *fine-grained relevance*.

MonoT5 jointly encodes each query–document pair, modeling interactions a bi-encoder cannot. Applied only to the top candidate set for tractability.

Late-stage relevance modeling captures nuanced financial reasoning — and amplifies the gains from chunking.

TOP-K BEFORE → AFTER MONOT5

#1	doc_4188 · "REIT vs. real estate stock – what to know"	▲ 4
#2	doc_0921 · "Diversification basics for retail investors"	▼ 1
#3	doc_7704 · "Dividend treatment of REIT distributions"	▲ 2
#4	doc_1156 · "How real estate IPOs are structured"	▼ 2
#5	doc_3329 · "Tax filing dates for K-1 issuers"	▼ 3

RESULTS

Each stage compounds the gain.

#	PIPELINE	MAP	NDCG@10	R@10	P@5
01	TF-IDF	0.208	0.247	0.298	0.121
02	BM25 (lexical baseline)	0.213	0.252	0.305	0.124
03	BGE dense	0.301	0.349	0.418	0.171
04	BGE + instruction prefix	0.327	0.378	0.446	0.183
05	Hybrid BM25 + BGE · RRF fusion ↓ DROP	0.265	0.308	0.371	0.151
06	Hybrid BM25 + BGE · linear fusion ↓ SLIGHT DROP	0.319	0.371	0.439	0.180
07	BGE + prefix + LLM rewriting & HyDE	0.345	0.402	0.481	0.194
08	+ Document chunking	0.371	0.428	0.512	0.207
09	+ MonoT5 re-ranking · winning configuration	0.412	0.471	0.547	0.226

FINDING Adding BM25 to a strong dense retriever hurt performance — RRF most of all. The wins came from prefixing, rewriting, chunking, and MonoT5, not from lexical fusion.

LESSONS

Three things worth remembering.

01

Read the model's *training paradigm*.

Prefixing queries with the right retrieval instruction unlocked a step-change in BGE performance.

02

Rewriting is *diversification*, not substitution.

Combining the original query with rewrites and HyDE drafts beat replacing it. Multiple semantic views cover ambiguous intent.

03

More signals isn't always *more signal*.

Adding BM25 to a strong dense retriever introduced noise. Hybrid wins came from chunking and re-ranking, not from fusion alone.

FUTURE WORK

From retrieval to *answers*.

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- 01 **Tune fusion weights systematically.**
 - 02 **Difficulty-aware re-ranking.**
 - 03 **Extend to retrieval-augmented generation.**
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The next step is a system that doesn't just rank documents — it answers questions, in plain language, for non-experts.

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*Clear access to financial
information helps people
make better life decisions.*

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